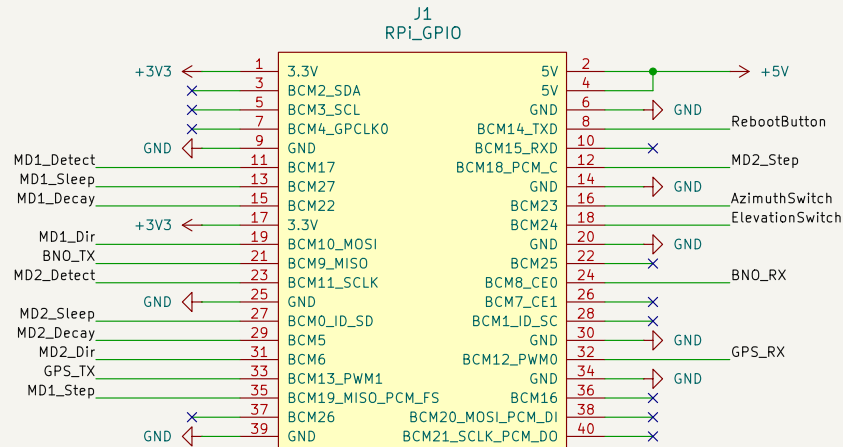
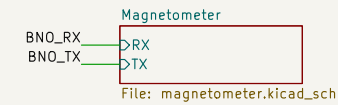


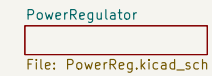
Raspberry Pi 4 GPIO Connector



BNO055 Accel/Gyro/Magno



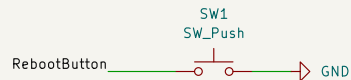
12v -> 5v Power Reg



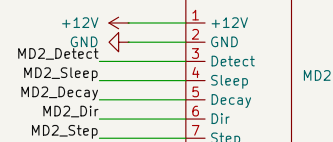
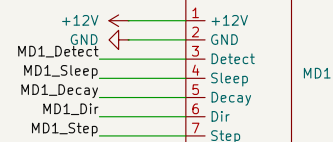
12v Input



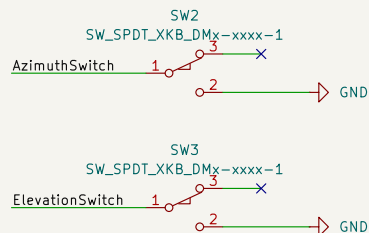
Reboot Button



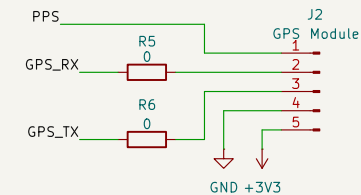
Motor Driver Connection



Motor Limit Switches



GPS Module Connector



Parts
R5-6: RMCF1206ZT0R00
GPS: ATGM336H

Notes
- Not all parts have been chosen yet
so some have been left blank.

Motor Driver by Lucas Moidel
Main HAT PCB Designed by Josh Carlson
Grilled Cheese Engineering

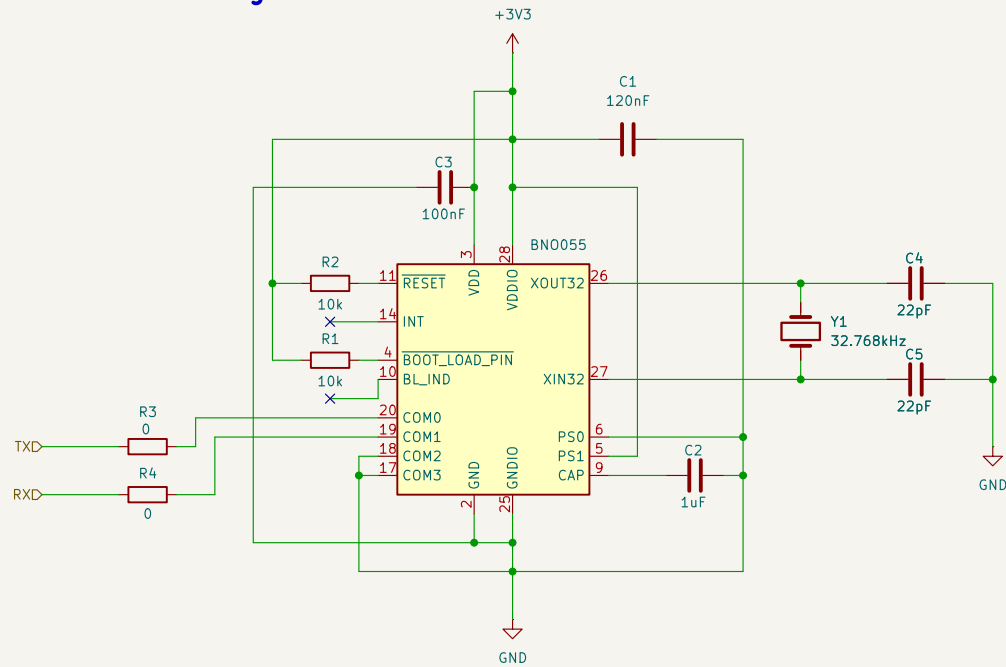
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Title: Antenna Rotor Hat Main Sheet

Size: A4 Date: 2026-09-06
KiCad E.D.A. 10.0.3

Rev: 0.1.0
Id: 1/4

12v to 5v Power Regulator



Parts:

- BNO055: BNO055
- C1: CC0805KRX7R9BB124
- C2: CL21F105ZBFNNNG
- C3: CL21B104KBCNNNC
- C4-5: CL21C220JBANNNC
- R1-2: RMC1206FT10K0
- R3-4: RMC1206ZT0R00
- Y1: AB38T-32.768KHZ

Motor Driver by Lucas Moidel
Main HAT PCB Designed by Josh Carlson

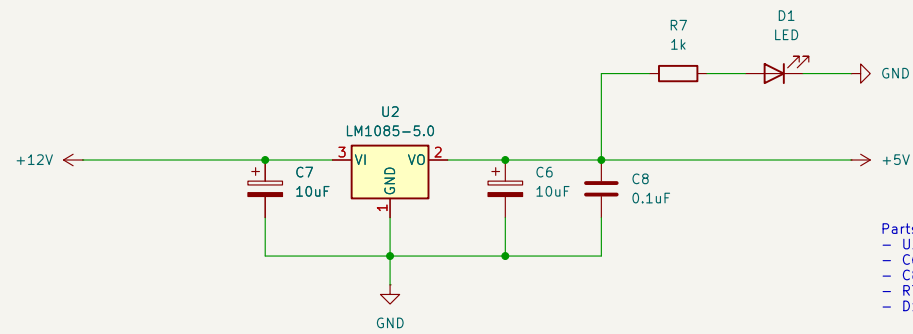
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Title:

Size: A4 Date: 2026-09-06
KiCad E.D.A. 10.0.3

Rev: 0.1.0
Id: 2/4

12v to 5v Power Regulator



Parts:

- U2: LM1085IT-5.0/NOPB
- C6-7: 50YXJ10M5X11
- C8: CL21B104KBCNNNC
- R7: CR1206-FX-1001ELF
- D1: LTST-C150KRKT

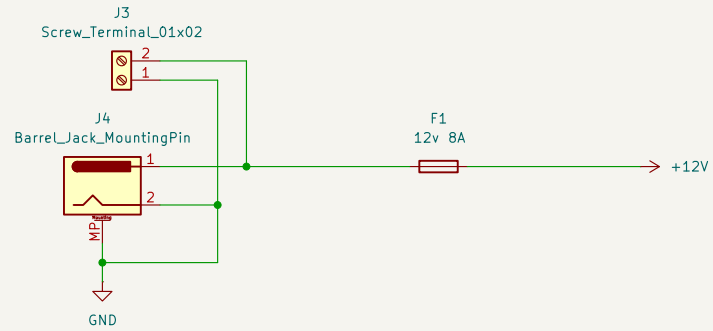
Motor Driver by Lucas Moidel
Main HAT PCB Designed by Josh Carlson

Sheet: /PowerRegulator/
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Title:

Size: A4	Date: 2026-09-06	Rev: 0.1.0
KiCad E.D.A. 10.0.3		Id: 3/4

Power Input Circuit



Notes:

- Supports 2 methods of input, barrel jack and screw terminal
- Fuse will trip if more than 8A is pulled through the circuit.
This is well above our estimated max current draw of 6A

Motor Driver by Lucas Moidel
Main HAT PCB Designed by Josh Carlson

Sheet: /PowerInput/
File: PowerIn.kicad_sch

Title:

Size: A4	Date: 2026-09-06	Rev: 0.1.0
KiCad E.D.A. 10.0.3		Id: 4/4